Answer Key — The Amazing World of Solutes, Solvents, and Solutions

Final answers with brief reasons.

Objective

Q1	**(b) solvent** — water does the dissolving — the solvent.
Q2	**(b) saturated** — that is a saturated solution.
Q3	**(a) increases** — most solids dissolve more in hotter water.
Q4	**(b) decreases** — gas solubility falls with temperature.
Q5	**(a) oxygen** — dissolved oxygen sustains aquatic life.
Q6	**(b) False** — sand forms a non-uniform mixture, not a solution.
Q7	**solubility** — that maximum is solubility.
Q8	**dilute** — little solute → dilute.

Short answer

Q9	Solute = the substance that dissolves (sugar); solvent = the one it dissolves in (water).
Q10	Solids: solubility increases with temperature. Gases: solubility decreases with temperature.
Q11	One that can dissolve no more solute at a given temperature; heating it increases solubility, so it can then dissolve more.

Long / application

Q12	Saturate water with baking soda (or sugar) at room temperature until some stays undissolved. Heat while stirring; the leftover dissolves. Add more until saturated, heat further — it dissolves again. This shows solubility of the solid increases with temperature.
Q13	Gas solubility decreases as temperature rises. Cold water holds more dissolved oxygen, so it supports more aquatic life; warm water holds less. A warm fizzy drink holds less dissolved carbon dioxide, so the gas escapes and it goes flat.

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